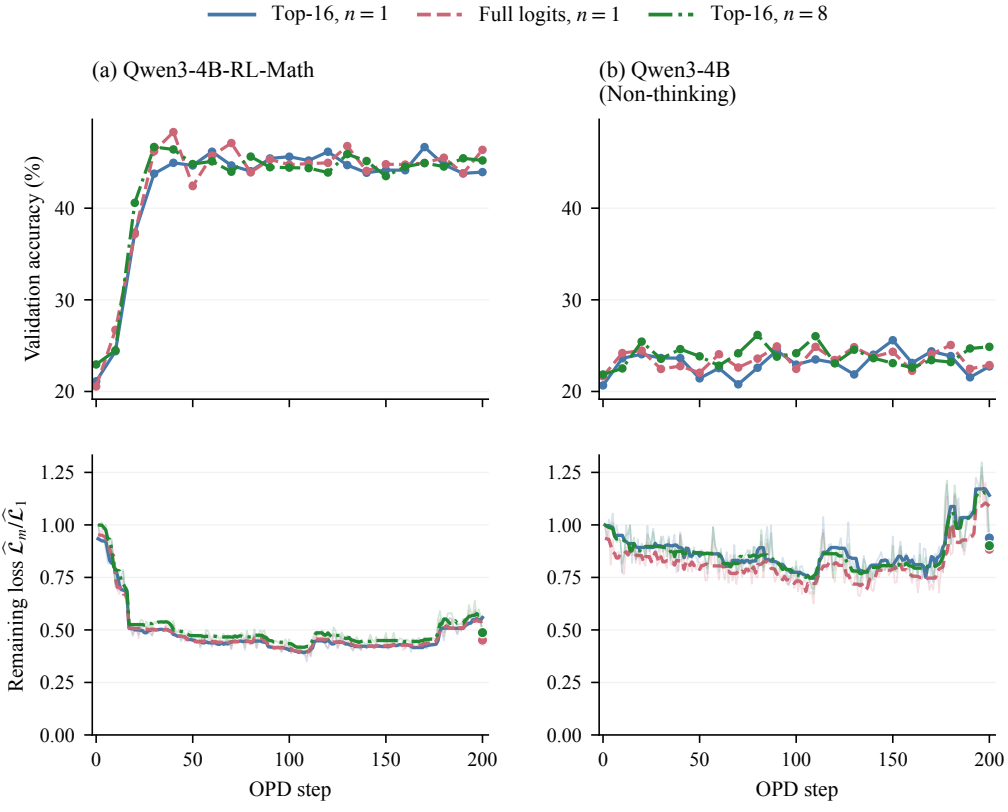

Top-16, $n = 1$
Full logits, $n = 1$
Top-16, $n = 8$
(a) Qwen3-4B-RL-Math
(b) Qwen3-4B (Non-thinking)
Validation accuracy (%)
Remaining loss $\widehat{\mathcal{L}}_m/\widehat{\mathcal{L}}_1$
OPD step
OPD step